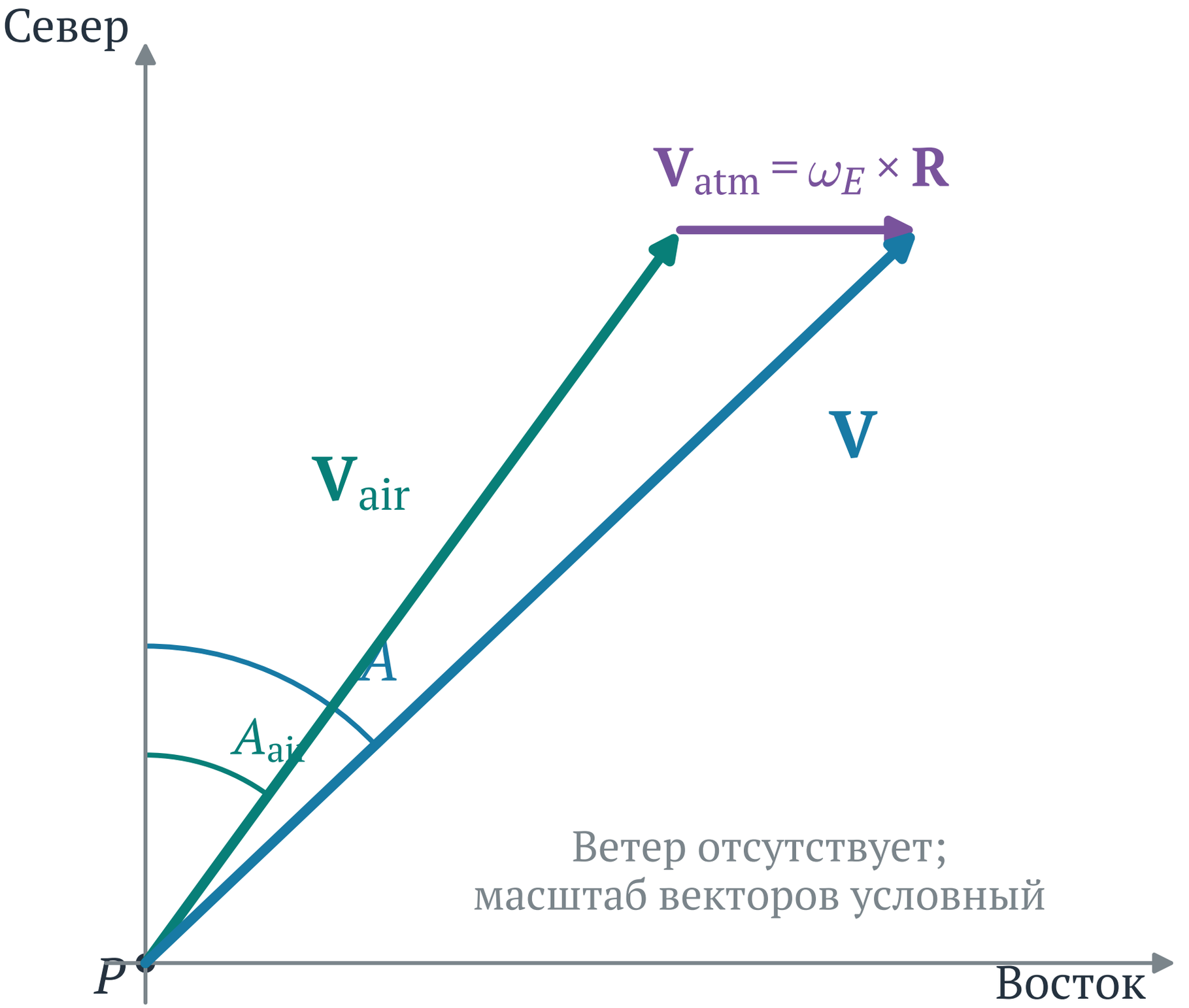

$$\mathbf{V} = \mathbf{V}_{\text{air}} + \omega_E \times \mathbf{R}$$